

Trigonometry & Additional Topics

ANSWER KEY



Right-triangle trigonometry

- 1 In the right triangle shown below, the hypotenuse is 20 and the side opposite angle θ is 12. Find the length of the adjacent side, then find $\sin \theta$, $\cos \theta$, and $\tan \theta$.

$$\text{Adjacent} = \sqrt{20^2 - 12^2} = \sqrt{400 - 144} = \sqrt{256} = 16. \sin \theta = \frac{12}{20} = \frac{3}{5}. \cos \theta = \frac{16}{20} = \frac{4}{5}.$$
$$\tan \theta = \frac{12}{16} = \frac{3}{4}.$$

- 2 In a right triangle, angle θ has adjacent side 7 and hypotenuse 25. Find $\sin \theta$ and $\tan \theta$.

$$\text{Opposite} = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24. \sin \theta = \frac{24}{25}. \tan \theta = \frac{24}{7}.$$

- 3 A 15-foot ladder leans against a wall, making a 50° angle with the ground. Find how high up the wall the ladder reaches, to the nearest tenth of a foot.

$$\text{Height} = 15 \sin(50^\circ) \approx 15(0.766) \approx 11.5 \text{ feet.}$$

The Pythagorean identity

- 4 Given $\sin \theta = \frac{5}{13}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- $$\cos^2 \theta = 1 - \frac{25}{169} = \frac{144}{169}. \text{ Since } \theta \text{ is acute, } \cos \theta = \frac{12}{13}.$$
- 5 Given $\cos \theta = \frac{8}{17}$ with θ acute, find $\sin \theta$.
- $$\sin^2 \theta = 1 - \frac{64}{289} = \frac{225}{289}. \text{ Since } \theta \text{ is acute, } \sin \theta = \frac{15}{17}.$$

Exact trigonometric values

- 6 State the exact value of $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{4}\right)$.
- $$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}. \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}.$$
- 7 State the exact value of $\tan(60^\circ)$ and $\sin(90^\circ)$.
- $$\tan(60^\circ) = \sqrt{3}. \sin(90^\circ) = 1.$$

Solving for missing sides and angles

- 8 A surveyor stands 150 feet from the base of a tower and measures the angle of elevation to the top as 32° , as shown below. Find the height of the tower, to the nearest foot.

$$h = 150 \tan(32^\circ) \approx 150(0.6249) \approx 94 \text{ feet.}$$

- 9 From the top of a 60-foot cliff, the angle of depression to a boat is 25° . Find the horizontal distance from the base of the cliff to the boat, to the nearest foot.

The angle of depression equals the angle of elevation from the boat, 25° , so $\tan(25^\circ) = \frac{60}{d}$, giving

$$d = \frac{60}{\tan(25^\circ)} \approx \frac{60}{0.4663} \approx 129 \text{ feet.}$$

Complex numbers: arithmetic

- 10 Add $(4 + 3i) + (2 - 7i)$.

$$6 - 4i.$$

- 11 Multiply $(3 + 2i)(1 - 4i)$.

$$3 - 12i + 2i - 8i^2 = 3 - 10i - 8(-1) = 11 - 10i.$$

- 12 Simplify i^7 , using $i^2 = -1$.

$$i^7 = i^4 \cdot i^3 = 1 \cdot i^3 = i^2 \cdot i = -1 \cdot i = -i.$$

Complex numbers: modulus and conjugates

- 13 Find the modulus $|6 - 8i|$.

$$\sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10.$$

- 14 Find the complex conjugate of $-3 + 5i$, and compute the product of the number and its conjugate.

$$\text{Conjugate: } -3 - 5i. \text{ Product} = (-3 + 5i)(-3 - 5i) = 9 - 25i^2 = 9 + 25 = 34.$$

Vectors: magnitude and operations

- 15 Vector $v = (3, 4)$. Find its magnitude.

$$|v| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

- 16 Given vectors $u = (2, -5)$ and $v = (-6, 3)$, find $u + v$.

$$u + v = (2 + (-6), -5 + 3) = (-4, -2).$$

- 17 Given vector $w = (8, -6)$, find $2w$ and the magnitude of $2w$.

$$2w = (16, -12). |2w| = \sqrt{16^2 + 12^2} = \sqrt{256 + 144} = \sqrt{400} = 20.$$

Radian measure

- 18 Convert 150° to radians, in terms of π .

$$150 \times \frac{\pi}{180} = \frac{5\pi}{6}.$$

- 19 Convert $\frac{7\pi}{4}$ radians to degrees.

$$\frac{7\pi}{4} \times \frac{180}{\pi} = 7 \times 45^\circ = 315^\circ.$$

A well-designed word problem: measuring the width of a river

- 20 This question has three parts. A surveyor stands at point A directly across a river from a tree at point B (so angle $A = 90^\circ$), then walks 80 meters along the riverbank to point C . (a) At point C , the angle between the bank and the line of sight to the tree is 35° . Use the tangent ratio to find AB , the width of the river, to the nearest meter. (b) Find the direct distance CB from C to the tree, to the nearest meter. (c) A colleague measures from a point 50 meters past C (so 130 meters from A) and finds the angle to the tree is 24° . Use this second measurement to find a second estimate of AB , to the nearest meter, and state which measurement gives the larger estimate.

(a) $\tan(35^\circ) = \frac{AB}{80}$, so $AB = 80 \tan(35^\circ) \approx 80(0.7002) \approx 56$ meters. (b) $CB = \frac{80}{\cos(35^\circ)} \approx \frac{80}{0.8192} \approx 98$ meters. (c) $AB = 130 \tan(24^\circ) \approx 130(0.4452) \approx 58$ meters. The second measurement (58 m) is slightly larger than the first (56 m); the two estimates are reasonably consistent.

Multiple choice — easy

- 21 In a right triangle, the hypotenuse is 87 and the side opposite angle θ is 60. Find $\sin \theta$.

- a. $\frac{21}{29}$
- b. $\frac{20}{29}$ ✓
- c. $\frac{20}{21}$
- d. $\frac{62}{87}$

$$\sin \theta = \frac{60}{87} = \frac{20}{29}.$$

22 In a right triangle, the side opposite angle θ is 6 and the side adjacent is 8. Find $\tan \theta$.

- a. 1
- b. $\frac{4}{5}$
- c. $\frac{3}{5}$
- d. $\frac{3}{4}$ ✓

$$\tan \theta = \frac{6}{8} = \frac{3}{4}.$$

23 In a right triangle, the hypotenuse is 50 and the side opposite angle θ is 14. Find $\sin \theta$.

- a. $\frac{8}{25}$
- b. $\frac{7}{25}$ ✓
- c. $\frac{24}{25}$
- d. $\frac{7}{24}$

$$\sin \theta = \frac{14}{50} = \frac{7}{25}.$$

24 In a right triangle, the side opposite angle θ is 24 and the side adjacent is 45. Find $\tan \theta$.

- a. $\frac{8}{17}$
- b. $\frac{15}{17}$
- c. $\frac{26}{45}$
- d. $\frac{8}{15}$ ✓

$$\tan \theta = \frac{24}{45} = \frac{8}{15}.$$

25 In a right triangle, the hypotenuse is 25 and the side adjacent to angle θ is 24. Find $\cos \theta$.

- a. $\frac{26}{25}$
- b. $\frac{24}{25}$ ✓
- c. $\frac{7}{24}$
- d. $\frac{7}{25}$

$$\cos \theta = \frac{24}{25}.$$

26 In a right triangle, the hypotenuse is 5 and the side opposite angle θ is 3. Find $\sin \theta$.

a. $\frac{4}{5}$

b. $\frac{3}{4}$

c. 1

d. $\frac{3}{5}$ ✓

$$\sin \theta = \frac{3}{5}.$$

27 In a right triangle, the side opposite angle θ is 40 and the side adjacent is 42. Find $\tan \theta$.

a. 1

b. $\frac{21}{29}$

c. $\frac{20}{29}$

d. $\frac{20}{21}$ ✓

$$\tan \theta = \frac{40}{42} = \frac{20}{21}.$$

28 In a right triangle, the hypotenuse is 41 and the side adjacent to angle θ is 40. Find $\cos \theta$.

a. $\frac{9}{40}$

b. $\frac{9}{41}$

c. $\frac{42}{41}$

d. $\frac{40}{41}$ ✓

$$\cos \theta = \frac{40}{41}.$$

- 29 In a right triangle, the hypotenuse is 15 and the side opposite angle θ is 9. Find $\sin \theta$.

- a. $\frac{3}{4}$
- b. $\frac{11}{15}$
- c. $\frac{4}{5}$
- d. $\frac{3}{5}$ ✓

$$\sin \theta = \frac{9}{15} = \frac{3}{5}.$$

- 30 In a right triangle, the side opposite angle θ is 18 and the side adjacent is 80. Find $\tan \theta$.

- a. $\frac{40}{41}$
- b. $\frac{9}{41}$
- c. $\frac{9}{40}$ ✓
- d. $\frac{1}{4}$

$$\tan \theta = \frac{18}{80} = \frac{9}{40}.$$

- 31 In a right triangle, the hypotenuse is 26 and the side adjacent to angle θ is 24. Find $\cos \theta$.

- a. $\frac{5}{12}$
- b. 1
- c. $\frac{12}{13}$ ✓
- d. $\frac{5}{13}$

$$\cos \theta = \frac{24}{26} = \frac{12}{13}.$$

- 32 In a right triangle, the hypotenuse is 51 and the side opposite angle θ is 24. Find $\sin \theta$.

- a. $\frac{15}{17}$
- b. $\frac{26}{51}$
- c. $\frac{8}{15}$
- d. $\frac{8}{17}$ ✓

$$\sin \theta = \frac{24}{51} = \frac{8}{17}.$$

33 In a right triangle, the hypotenuse is 29 and the side opposite angle θ is 20. Find $\sin \theta$.

a. $\frac{21}{29}$

b. $\frac{20}{29}$ ✓

c. $\frac{22}{29}$

d. $\frac{20}{21}$

$$\sin \theta = \frac{20}{29}.$$

34 In a right triangle, the hypotenuse is 10 and the side adjacent to angle θ is 8. Find $\cos \theta$.

a. 1

b. $\frac{4}{5}$ ✓

c. $\frac{3}{4}$

d. $\frac{3}{5}$

$$\cos \theta = \frac{8}{10} = \frac{4}{5}.$$

35 In a right triangle, the side opposite angle θ is 36 and the side adjacent is 105. Find $\tan \theta$.

a. $\frac{12}{35}$ ✓

b. $\frac{38}{105}$

c. $\frac{12}{37}$

d. $\frac{35}{37}$

$$\tan \theta = \frac{36}{105} = \frac{12}{35}.$$

Multiple choice — medium

36 Given $\sin \theta = \frac{11}{61}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{62}{61}$

b. $\frac{60}{61}$ ✓

c. 1

d. $\frac{11}{61}$

$$\cos^2 \theta = 1 - \frac{121}{3721} = \frac{3600}{3721}. \text{ Since } \theta \text{ is acute, } \cos \theta = \frac{60}{61}.$$

37 Given $\sin \theta = \frac{12}{37}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{35}{36}$

b. $\frac{12}{37}$

c. $\frac{36}{37}$

d. $\frac{35}{37}$ ✓

$$\cos^2 \theta = 1 - \frac{144}{1369} = \frac{1225}{1369}. \text{ Since } \theta \text{ is acute, } \cos \theta = \frac{35}{37}.$$

38 Given $\sin \theta = \frac{9}{41}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{40}{41}$ ✓

b. $\frac{9}{41}$

c. 1

d. $\frac{42}{41}$

$$\cos^2 \theta = 1 - \frac{81}{1681} = \frac{1600}{1681}. \text{ Since } \theta \text{ is acute, } \cos \theta = \frac{40}{41}.$$

39 Given $\sin \theta = \frac{5}{13}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{14}{13}$

b. 1

c. $\frac{12}{13}$ ✓

d. $\frac{5}{13}$

$$\cos^2 \theta = 1 - \frac{25}{169} = \frac{144}{169}. \text{ Since } \theta \text{ is acute, } \cos \theta = \frac{12}{13}.$$

40 Given $\cos \theta = \frac{21}{29}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{20}{29}$ ✓

b. $\frac{22}{29}$

c. $\frac{21}{29}$

d. $\frac{5}{7}$

$$\sin^2 \theta = 1 - \frac{441}{841} = \frac{400}{841}. \text{ Since } \theta \text{ is acute, } \sin \theta = \frac{20}{29}.$$

41 Given $\cos \theta = \frac{15}{17}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{9}{17}$

b. $\frac{1}{2}$

c. $\frac{8}{17}$ ✓

d. $\frac{15}{17}$

$$\sin^2 \theta = 1 - \frac{225}{289} = \frac{64}{289}. \text{ Since } \theta \text{ is acute, } \sin \theta = \frac{8}{17}.$$

42 Given $\cos \theta = \frac{60}{61}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{11}{60}$

b. $\frac{60}{61}$

c. $\frac{12}{61}$

d. $\frac{11}{61}$ ✓

$$\sin^2 \theta = 1 - \frac{3600}{3721} = \frac{121}{3721}. \text{ Since } \theta \text{ is acute, } \sin \theta = \frac{11}{61}.$$

- 43 Given $\sin \theta = \frac{7}{25}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

- a. $\frac{7}{25}$
- b. $\frac{26}{25}$
- c. 1
- d. $\frac{24}{25}$ ✓

$$\cos^2 \theta = 1 - \frac{49}{625} = \frac{576}{625}. \text{ Since } \theta \text{ is acute, } \cos \theta = \frac{24}{25}.$$

- 44 Given $\cos \theta = \frac{35}{37}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

- a. $\frac{12}{37}$ ✓
- b. $\frac{35}{37}$
- c. $\frac{13}{37}$
- d. $\frac{1}{3}$

$$\sin^2 \theta = 1 - \frac{1225}{1369} = \frac{144}{1369}. \text{ Since } \theta \text{ is acute, } \sin \theta = \frac{12}{37}.$$

- 45 Given $\cos \theta = \frac{4}{5}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

- a. $\frac{4}{5}$
- b. 1
- c. $\frac{3}{5}$ ✓
- d. $\frac{3}{4}$

$$\sin^2 \theta = 1 - \frac{16}{25} = \frac{9}{25}. \text{ Since } \theta \text{ is acute, } \sin \theta = \frac{3}{5}.$$

- 46 Given $\sin \theta = \frac{20}{29}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

- a. $\frac{22}{29}$
- b. $\frac{20}{29}$
- c. $\frac{21}{29}$ ✓
- d. $\frac{3}{4}$

$$\cos^2 \theta = 1 - \frac{400}{841} = \frac{441}{841}. \text{ Since } \theta \text{ is acute, } \cos \theta = \frac{21}{29}.$$

47 Given $\cos \theta = \frac{12}{13}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{12}{13}$

b. $\frac{5}{12}$

c. $\frac{5}{13}$ ✓

d. $\frac{6}{13}$

$$\sin^2 \theta = 1 - \frac{144}{169} = \frac{25}{169}. \text{ Since } \theta \text{ is acute, } \sin \theta = \frac{5}{13}.$$

48 Given $\sin \theta = \frac{3}{5}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{4}{5}$ ✓

b. $\frac{3}{5}$

c. $\frac{6}{5}$

d. 1

$$\cos^2 \theta = 1 - \frac{9}{25} = \frac{16}{25}. \text{ Since } \theta \text{ is acute, } \cos \theta = \frac{4}{5}.$$

49 Given $\cos \theta = \frac{40}{41}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

a. $\frac{10}{41}$

b. $\frac{40}{41}$

c. $\frac{9}{41}$ ✓

d. $\frac{9}{40}$

$$\sin^2 \theta = 1 - \frac{1600}{1681} = \frac{81}{1681}. \text{ Since } \theta \text{ is acute, } \sin \theta = \frac{9}{41}.$$

50 Given $\cos \theta = \frac{24}{25}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

- a. $\frac{8}{25}$
- b. $\frac{24}{25}$
- c. $\frac{7}{25}$ ✓
- d. $\frac{7}{24}$

$$\sin^2 \theta = 1 - \frac{576}{625} = \frac{49}{625}. \text{ Since } \theta \text{ is acute, } \sin \theta = \frac{7}{25}.$$

Multiple choice — hard

51 Multiply $(8 - i)(1 + 5i)$.

- a. $16 + 39i$
- b. $13 + 39i$ ✓
- c. $3 + 39i$
- d. $13 + 41i$

$$8 + 40i - i - 5i^2 = 8 + 39i - 5(-1) = 13 + 39i.$$

52 Multiply $(1 - 3i)(9 - 9i)$.

- a. $-18 + 18i$
- b. $-18 - 34i$
- c. $36 - 36i$
- d. $-18 - 36i$ ✓

$$9 - 9i - 27i + 27i^2 = 9 - 36i + 27(-1) = -18 - 36i.$$

53 Multiply $(-1 - 9i)(5 + 2i)$.

a. $13 - 47i$ ✓

b. $-23 - 47i$

c. $13 + 43i$

d. $13 - 45i$

$$-5 - 2i - 45i - 18i^2 = -5 - 47i - 18(-1) = 13 - 47i.$$

54 Multiply $(6 - 6i)(-2 - 8i)$.

a. $-60 - 36i$ ✓

b. $36 - 36i$

c. $-60 - 34i$

d. $-60 - 60i$

$$-12 - 48i + 12i + 48i^2 = -12 - 36i + 48(-1) = -60 - 36i.$$

55 Multiply $(7 + 3i)(-1 - 8i)$.

a. $-31 - 59i$

b. $17 - 59i$ ✓

c. $17 - 53i$

d. $17 - 57i$

$$-7 - 56i - 3i - 24i^2 = -7 - 59i - 24(-1) = 17 - 59i.$$

56 Given vectors $u = (0, 5)$ and $v = (7, 0)$, find $u + v$.

a. $(7, 3)$

b. $(9, 5)$

c. $(5, 7)$

d. $(7, 5)$ ✓

$$u + v = (0 + 7, 5 + 0) = (7, 5).$$

57 Given vectors $u = (9, -8)$ and $v = (5, -3)$, find $u + v$.

a. $(-11, 14)$

b. $(14, -11)$ ✓

c. $(14, -13)$

d. $(16, -11)$

$$u + v = (9 + 5, (-8) + (-3)) = (14, -11).$$

58 Given vectors $u = (4, 9)$ and $v = (1, -7)$, find $u + v$.

a. $(2, 5)$

b. $(5, 2)$ ✓

c. $(5, 0)$

d. $(7, 2)$

$$u + v = (4 + 1, 9 + (-7)) = (5, 2).$$

59 Given vectors $u = (3, -1)$ and $v = (-8, -5)$, find $u + v$.

a. $(-5, -6)$ ✓

b. $(-3, -6)$

c. $(-5, -8)$

d. $(-6, -5)$

$$u + v = (3 + (-8), (-1) + (-5)) = (-5, -6).$$

60 Given vectors $u = (-4, -2)$ and $v = (0, 5)$, find $u + v$.

- a. $(-4, 1)$
- b. $(3, -4)$
- c. $(-2, 3)$
- d. $(-4, 3)$ ✓

$$u + v = ((-4) + 0, (-2) + 5) = (-4, 3).$$

61 Convert 90° to radians, in terms of π .

- a. π
- b. $\frac{3\pi}{2}$
- c. $\frac{\pi}{3}$
- d. $\frac{\pi}{2}$ ✓

$$90 \times \frac{\pi}{180} = \frac{\pi}{2}.$$

62 Convert 120° to radians, in terms of π .

- a. $\frac{4\pi}{3}$
- b. $\frac{\pi}{2}$
- c. $\frac{2\pi}{3}$ ✓
- d. π

$$120 \times \frac{\pi}{180} = \frac{2\pi}{3}.$$

63 Convert 75° to radians, in terms of π .

- a. $\frac{5\pi}{6}$
- b. $\frac{\pi}{2}$
- c. $\frac{5\pi}{13}$
- d. $\frac{5\pi}{12}$ ✓

$$75 \times \frac{\pi}{180} = \frac{5\pi}{12}.$$

64 Convert 345° to radians, in terms of π .

- a. $\frac{23\pi}{13}$
- b. $\frac{23\pi}{6}$
- c. $\frac{23\pi}{12}$ ✓
- d. 2π

$$345 \times \frac{\pi}{180} = \frac{23\pi}{12}.$$

65 Convert 180° to radians, in terms of π .

- a. π ✓
- b. 3π
- c. $\frac{\pi}{2}$
- d. 2π

$$180 \times \frac{\pi}{180} = \pi.$$